Benjamin P. Gregory

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EDUCATION

University of Maryland, College Park Expected May 2027

Ph.D. Candidate, Entomology (advisor: Megan Fritz)

- Dissertation: Thermal adaptation and hybridization in *Culex* mosquitoes: Implications for urban West Nile virus transmission

Vassar College May 2020

B.A., Biology; Anthropology minor. GPA 3.73/4.00. Departmental and Institutional Honors

- Honors thesis: Low-frequency background noise limits the influence of the audience effect on male *Dendropsophus ebraccatus* calling behavior

PUBLICATIONS

Gregory, B. P., Aronson, H., Vinauger, C., Lahondère, C., & Fritz, M. L. (2026). Stage-specific divergence in larval thermal tolerance among populations of the *Culex pipiens* Assemblage. *Journal of Medical Entomology*, 63(4), tjag115. https://doi.org/10.1093/jme/tjag115

Fuchsman, C. A., Duffy, M. E., Cram, J. A., Huanca-Valenzuela, P., **Gregory, B. P.**, Plough, L. V., Pierson, J. J., Fitzgerald, C. L., Devol, A. H., & Keil, R. G. (2025). Contributions of vertically migrating metazoans to sinking and suspended particulate matter fuel N2 production in the eastern tropical North Pacific oxygen deficient zone. *Global Biogeochemical Cycles*, 39(3), e2024GB008365. https://doi.org/10.1029/2024GB008365

Huang, C. S., Legett, H. D., Plough, L. V., Aguilar, R., Fitzgerald, C., **Gregory, B.**, Heggie, K., Lee, B., Richie, K. D., Harbold, W., & Ogburn, M. B. (2023). Early detection and recovery of river herring spawning habitat use in response to a mainstem dam removal. *PLOS ONE*, 18(5), e0284561. https://doi.org/10.1371/journal.pone.0284561

Under review and in preparation

Gregory, B. P.*, Arsenault-Benoit, A.*, Irwin, P., Price, K. J., Witmier, B. J., Fitzpatrick, M. C., & Fritz, M. L. (under review). Climate and urbanization structure West Nile virus vector communities: Tools for surveillance under environmental change. *Ecological Applications*. (*Co-first authors.) https://doi.org/10.64898/2026.07.24.738033.

Gregory, B. P., Arsenault-Benoit, A., Bell, K. L., & Fritz, M. L. (2026). Urban enrichment of *molestus* ancestry and opposing temperature clines in a metropolitan *Culex pipiens* hybrid zone. *bioRxiv*. https://doi.org/10.64898/2026.10.30.763117. Submitted to Molecular Ecology.

FELLOWSHIPS, HONORS, AND AWARDS

Ann G. Wylie Dissertation Fellowship, University of Maryland	2026
Third place, Entomological Society of Washington Student Poster Competition	2026
Third place, Virginia Mosquito Control Association Student Poster Competition	2026
Charles Reichelderfer Memorial Mentorship Award	2025
Teaching Achievement Award	2024–2025
First place, ESA-MUVE National Student Poster Competition	2023
CMNS Dean's Fellowship Award, University of Maryland	2022
Member, Sigma Xi Scientific Research Honor Society	2020
Virginia Swinburne Brownell Prize for Excellence in Biology, Vassar College	2020
Kate Roberts Prize for Excellence in Biology, Vassar College	2018

RESEARCH EXPERIENCE

Graduate Researcher, Department of Entomology, University of Maryland 2022–Present
Advisor: Megan Fritz

- Showed that larval thermal tolerance differs among *Culex pipiens* Assemblage populations sampled along a latitudinal and temperature gradient in the eastern United States, and that the differences vary by life stage (Gregory et al. 2026).
- Contributed to a long-term field study of *Culex* West Nile virus vectors along urbanization gradients in the eastern United States; modeled how landscape features and phenology shape relative *Cx. pipiens* and *Cx. restuans* abundance.
- Whole-genome Pool-seq study of the *Cx. pipiens* hybrid zone across the Baltimore–Washington corridor (24 field pools, 3 reference pools): ancestry, genotype–environment association, and candidate loci along temperature and urbanization gradients.

Faculty Research Assistant, Environmental Genomics, UMCES Horn Point Laboratory 2020–2022

- Environmental DNA sampling, quantitative PCR, and genetic assay development; population genetic analysis of estuarine and marine animals; tested a novel CRISPR-based eDNA sampling method.

Evolutionary Ecology Research Assistant, Vassar College (Touchon Lab) 2018–2020

- Studies of animal behavior, phenotypic plasticity, and physiology. Undergraduate Research Summer Institute (URSI) Fellow, Summer 2019: behavioral mate-preference study in a colony of *Dendropsophus ebraccatus* treefrogs.

SIT Panama: Tropical Ecology, Marine Ecosystems, and Biodiversity Conservation Spring 2019

- Field ecology methods; independent study of great-tailed grackle (*Quiscalus mexicanus*) vocalization pitch and traffic noise along the Cinta Costera highway and in downtown Gamboa, using field recordings and acoustic analysis (Gregory 2019, SIT Independent Study Project Collection 3115).

Ecological Preserve Land Steward and Research Assistant, Vassar College Summer 2018

- Maintained trails, collected data for long-term research projects, and managed invasive species.

TEACHING EXPERIENCE

Guest Lecturer, BSCI483 Insects, Pathogens, and Public Health, University of Maryland Oct 2026

- "Vector-borne disease in a changing climate" (course instructor: Megan Fritz).

Course Development, BSCI483 Insects, Pathogens, and Public Health, University of Maryland 2026

- Developed lab activities teaching students to identify the cryptic species *Cx. pipiens* and *Cx. restuans* by PCR, the assay used in my dissertation research.

Teaching Assistant, Principles of Genetics, University of Maryland 2026

Teaching Assistant, Introduction to Entomology, University of Maryland 2025

- Led weekly discussion sections, planned lectures, graded assignments.

Teaching Assistant, BSCI483 Insects, Pathogens, and Public Health, University of Maryland 2023–2026

- Led two weekly discussion sections, graded assignments, and supported students with course material.

Academic Intern, Introductory Biology and Evolutionary Genetics, Vassar College 2018–2020

- Prepared and guided laboratory and field activities and assisted students with material and lab methods.

MENTORING EXPERIENCE

Mentor, UMD Student Research in the Fritz Lab Summer 2023–Present

- Trained and supervised undergraduates on laboratory and field projects.

Undergraduate Team Science Research Project Mentor Spring 2025, 2026

- Led a team of six undergraduates through a semester-long project on the phenology and thermal tolerance of *Cx. pipiens* and *Cx. restuans* (2025).

Summer Opportunities in Agriculture and the Environment (SOARE) Program Mentor Summer 2024

- Mentored an undergraduate through a project on larval cold tolerance in *Cx. pipiens* and *Cx. quinquefasciatus*, from design to presentation.

Northeast Regional Vector-Borne Disease Teaching & Evaluation Center Mentor Summer 2024

- Mentored an undergraduate in field and laboratory techniques for *Culex* mosquito monitoring.

ORAL PRESENTATIONS

Gregory B, Arsenault-Benoit A, Bell K, Fritz M (October 2026) How warming and urbanization are reshaping insecticide-resistance genes and lineage composition in mid-Atlantic *Culex pipiens* mosquitoes. Pennsylvania Vector Control Association Annual Meeting, State College, PA.

Gregory B, Arsenault-Benoit A, Irwin P, Price K, Witmier B, Fitzpatrick M, Fritz M (April 2026) Landscape features and phenology differentially influence *Culex pipiens* and *Culex restuans* abundance in Eastern North America. Midwest Center of Excellence in Vector-Borne Disease Meeting, remote.

Gregory B, Fritz M (March 2026) Genomic evidence for climate-mediated introgression in a shifting hybrid zone. UMD Entomology colloquium series, College Park, MD.

Gregory B, Arsenault-Benoit A, Fitzpatrick M, Fritz M (January 2026) Landscape features and phenology influence relative *Culex* abundance along urbanization gradients in Maryland. Virginia Mosquito Control Association Annual Meeting, Virginia Beach, VA.

Gregory B (October 2025) Natural mosquito control for gardeners. Extension talk, Montgomery County Master Gardener Fall Conference, remote.

Gregory B, Arsenault-Benoit A, Fitzpatrick M, Fritz M (August 2025) Landscape features and phenology influence relative *Culex* abundance along urbanization gradients in Maryland. UMD Entomology Department annual retreat, College Park, MD.

Gregory B, Arsenault-Benoit A, Fitzpatrick M, Fritz M (July 2025) Landscape features and phenology influence relative *Culex* abundance along urbanization gradients in Maryland. Mid-Atlantic Zoonotic and Vector-Borne Disease Inter-Agency Workgroup, remote.

Gregory B, Fritz M (March 2025) Evidence of local adaptation to temperature in *Culex pipiens* Assemblage larvae. Entomological Society of America Eastern Branch meeting, Harrisburg, PA.

Gregory B, Fritz M (November 2024) Evidence of local adaptation to temperature in *Culex pipiens* Assemblage larvae. UMD Entomology colloquium series, College Park, MD.

Gregory B, Fritz M (January 2024) Turning up the heat: A comparison of larval thermal tolerance between *Culex pipiens* and *Culex quinquefasciatus*. Mid-Atlantic Mosquito Control Association conference, Annapolis, MD.

Gregory B, Fritz M (November 2023) Turning up the heat: A comparison of larval thermal tolerance between *Culex pipiens* and *Culex quinquefasciatus*. UMD Entomology colloquium series, College Park, MD.

POSTER PRESENTATIONS

Gregory B, Fritz M (May 2026) Introgression of adaptive alleles in the *Culex pipiens–quinquefasciatus* hybrid zone. Entomological Society of Washington Annual Student Poster Competition, Washington, DC. *Third place.*

Gregory B, Arsenault-Benoit A, Fitzpatrick M, Fritz M (January 2026) Landscape features and phenology influence relative *Culex* abundance along urbanization gradients in Maryland. Virginia Mosquito Control Association Annual Meeting, Virginia Beach, VA. *Third place, student competition.*

Gregory B, Arsenault-Benoit A, Fitzpatrick M, Fritz M (November 2025) Landscape features and phenology influence relative cryptic *Culex* abundance along an urbanization gradient in Eastern North America. Entomological Society of America Annual Meeting, Portland, OR.

Gregory B, Arsenault-Benoit A, Fitzpatrick M, Fritz M (November 2025) Landscape features and phenology influence relative cryptic *Culex* abundance along an urbanization gradient in Eastern North America. AAAS C. V. Riley Memorial Lecture, Washington, DC.

Gregory B, Arsenault-Benoit A, Fitzpatrick M, Fritz M (August 2025) Landscape features and phenology influence relative cryptic *Culex* abundance along an urbanization gradient in Eastern North America. Ecological Society of America Annual Meeting, Baltimore, MD.

Penafiel I, **Gregory B**, Fritz M (March 2025) Knocked out cold: Comparing larval tolerance of freezing temperatures between *Culex pipiens* and *Culex quinquefasciatus*. UMD AGNR Climate Adaptation and Mitigation Workshop, College Park, MD.

Gregory B, Fritz M (November 2023) Turning up the heat: A comparison of larval thermal tolerance between *Culex pipiens* and *Culex quinquefasciatus*. Entomological Society of America Annual Meeting, National Harbor, MD. *First place, ESA-MUVE student competition.*

OUTREACH AND SERVICE

Skype a Scientist participant (virtual and in-person K-12 classroom visits) 2022–Present

Treasurer, Entomology Student Organization, University of Maryland 2023–2024

TECHNICAL SKILLS

Computing: R (advanced; 8+ years), bash/shell scripting, Linux and HPC clusters, Python (working)

Population and landscape genomics: whole-genome Pool-seq analysis, genotype–environment association (BayPass), f3 admixture statistics, FST decomposition and ancestry inference, read-depth copy-number inference, structural-variant and inversion detection (DELLY)

Statistical modeling: negative-binomial GLMMs and mixed models, AIC-based model selection and likelihood-ratio testing, Gradient Forest compositional-turnover analysis

Geospatial and climate data: landscape-predictor extraction (impervious surface, tree canopy, NDVI, elevation), PRISM climate data, accumulated degree-day modeling

Wet lab and field: mosquito colony rearing, thermal-tolerance assays, field collection, DNA extraction for Pool-seq, environmental DNA sampling and qPCR

Other field methods: bird mist-netting, measuring, and banding (short course, Gamboa, Panama, 2019; owl-banding course, Vassar College, Fall 2019); acoustic analysis (Raven Pro)

Certifications and languages: IACUC certification for work with amphibians and fish; SIELE-certified bilingual in Spanish